

VINEET JETHANI

+91-7600446877 ◊ Surat, Gujarat ◊ heyvineetjethani@gmail.com ◊ [LinkedIn](#)

EDUCATION

Sardar Vallabhbhai National Institute Of Technology
B.Tech in Electronics and Communication Engineering – 8.27 CGPA

Surat, Gujarat
Aug 2023 – Expected July 2027

Senior Secondary (CBSE) – Score: 93%

Surat, Gujarat — 2023

Secondary (CBSE) – Score: 96.33%

Surat, Gujarat — 2021

EXPERIENCE

FPGA Developer Intern

May 2026 – Present

Vicharak Computers, Surat

- Completed an IR Transmitter–Receiver on the Shrike Lite FPGA (NEC protocol): 38 kHz carrier generator, PWM burst-frame encoder, and receiver demodulation FSM in Verilog.
- Designed NEC-compliant frame sequencing (leader code, address, command, stop-bit); verified TX–RX loopback on hardware with waveform-level timing analysis.
- Configured Renesas RA-series MCU via *Renesas Go Configure*; programmed MicroPython in *Thonny IDE* for peripheral interfacing and system-level validation.

ACADEMIC PROJECTS

EEG-Based Seizure Prediction using Spiking Neural Networks

- Processed EEG signals (0.5–40 Hz) via window-based segmentation and Butterworth filtering; implemented hybrid CNN + MLP classifier with spike-driven LIF neuron gating for efficient inference.
- Achieved **92.35%** accuracy, **92.38%** specificity, **85.82%** sensitivity (LOPO-CV); deployed on NVIDIA Jetson Nano at **151.53 μ J/inference**.

IR Transmitter–Receiver System – Shrike Lite FPGA (NEC Protocol)

- Implemented NEC TX/RX pipeline in Verilog: 38 kHz carrier, PWM burst encoder (leader code, 8-bit address/command, stop-bit), and timer-based FSM for receiver demodulation; validated loopback on hardware.

UART Transmitter

- Designed a UART Transmitter in Verilog with internal baud rate generator; 4-state FSM (IDLE, START, DATA, STOP) for 8N1 framing with LSB-first serialization.
- Verified via self-checking testbench in Xilinx Vivado; waveform-level timing analysis to validate setup/hold margins.

2G GSM Mobile Signal Jammer

- Designed a 900 MHz RF circuit: NE555 astable multivibrator driving a BF495 RF transistor; LC tank oscillator tuned to the 880–960 MHz GSM band via trimmer capacitors.
- Verified resonant frequency ($f = \frac{1}{2\pi\sqrt{LC}}$) via LTspice; analyzed Q-point stability (V_{CE} , I_C) for linear operation and maximum RF output power.

SKILLS

Area of Interest

Digital VLSI Design, FPGA Design, Embedded Systems, DSP

Core Subjects

Digital Electronics, CMOS VLSI Design, Computer Architecture

Simulation & Design Tools

Vivado, Cadence Virtuoso, LTspice, KiCAD, Renesas Go Configure

Hardware Description Languages

Verilog, VHDL

Protocols & Interfaces

NEC IR Protocol, UART, SPI

Programming

Python, MicroPython, C/C++ (basic)